

Building a Pose Map

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2. Building a Raster Map

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1. Building Raster Maps and Pose Maps using slam_toolbox

Note

- At least one raster map is required before the road network .geojson file.
- The pose map is used for fast relocalization during waypoint marking and navigation, improving the accuracy of global localization and thus the accuracy of waypoint marking.
- This demonstration uses the **Orin board**.

2. Building a Raster Map

Open ~/.bashrc, uncomment this line, and change it as shown in the image. (After learning about road network planning, please restore this comment to avoid affecting previous examples.)

```
vim ~/.bashrc
:wq #Save and exit
source ~/.bashrc #Update environment
```

```
jetson@yahboom: ~  
jetson@yahboom: ~ 118x36  
fl  
fl  
  
export LIBV4L_LOG_LEVEL=0  
export PATH=/usr/local/cuda-12.6/bin:$PATH  
export LD_LIBRARY_PATH=/usr/local/cuda-12.6/lib64:$LD_LIBRARY_PATH  
export LD_LIBRARY_PATH=/usr/local/lib:$LD_LIBRARY_PATH  
export PYTHONPATH=/usr/local/lib/python3.10/site-packages/:$PYTHONPATH  
  
export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp  
# unset RMW_IMPLEMENTATION  
export ROS_DOMAIN_ID=62  
export ROBOT_TYPE=A1  
export RPLIDAR_TYPE=tmnl      # c1, tmnl  
export CAMERA_TYPE=nuwa      # usb, nuwa  
  
export INIT_SERVO_S1=90      # 0~180  
export INIT_SERVO_S2=30      # 0~100  
  
source /opt/ros/humble/setup.bash  
source /home/jetson/yahboomcar_ros2_ws/software/library_ws/install/setup.bash  
source /home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/install/setup.bash  
  
# Read the number of rows and columns from the terminal size  
read -r rows cols < <(&stty size)  
  
# Define the system information text  
SystemInfo="System Information"  
# Calculate the length of the system information text  
SystemInfoLength=${#SystemInfo}  
# Calculate half of the number of columns in the terminal  
half_cols=$((cols / 2))  
# Calculate the padding needed to center the text horizontally
```

💡 Tip

- Before building the map, you can place some obstacles outside the roads on the map so that the actual position of the car can be referenced during navigation.
- Start slam_toolbox to build the map,

```
ros2 launch yahboomcar_nav map_cartographer_launch.py
```

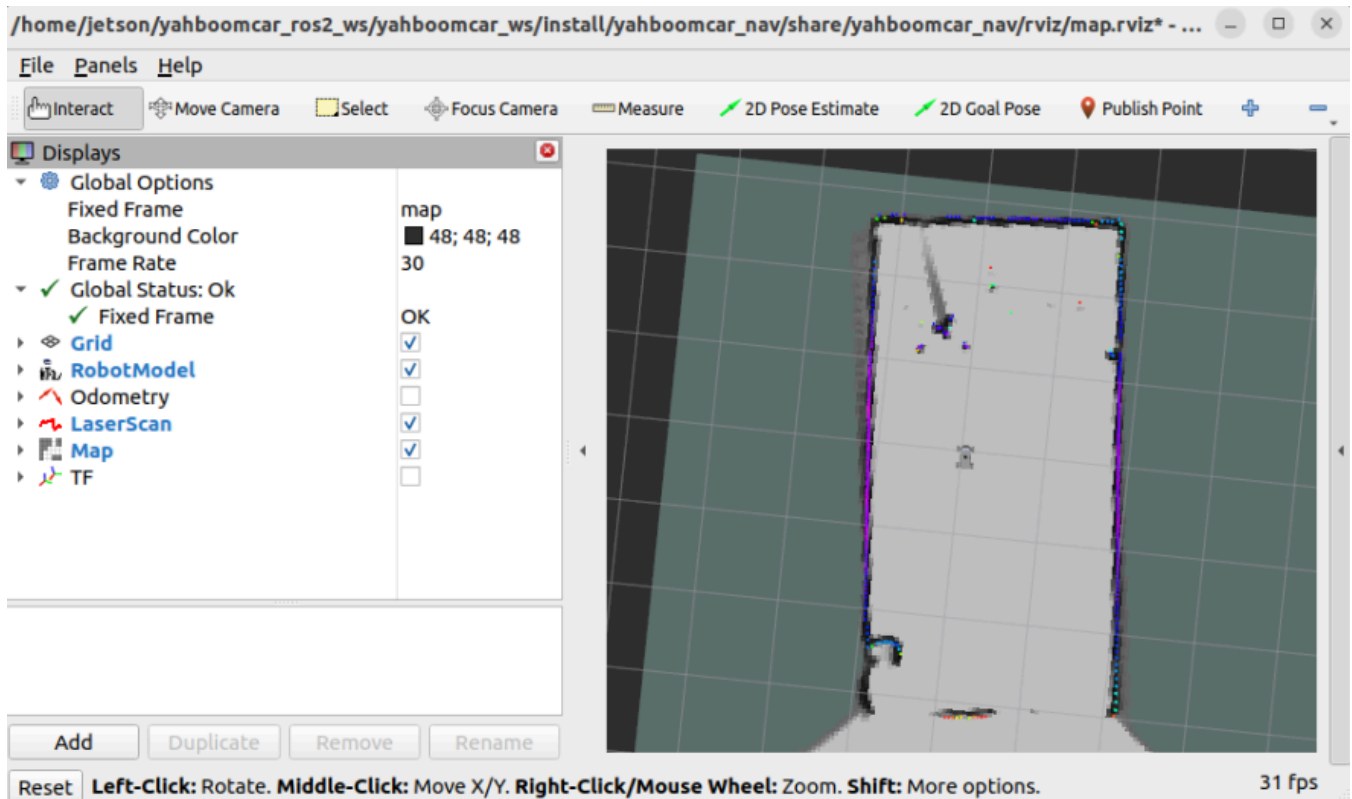
- To launch the rviz visualization, it is recommended to operate on a virtual machine using RDK X5.

```
ros2 launch yahboomcar_nav display_map_launch.py
```

- Launch the keyboard control node (a gamepad can also be used),

```
ros2 run yahboomcar_ctrl yahboom_keyboard
```

- Control the car's movement to build the map



- Save the raster map

```
#ORIN
ros2 launch yahboomcar_nav roda_save_map_launch.py map_name:=map

#RDK
ros2 launch yahboomcar_nav roda_save_map_launch.py map_name:=map
```

💡 Tip

- map_name specifies the **map name** for the saved raster map. If no startup parameter is specified, the default map name is `map`. All map files are saved in the `~/map` directory.

The terminal message `Map saved successfully` indicates that the map was saved successfully.

```
jetson@yahboom: ~  
jetson@yahboom: ~ 97x28  
-----  
jetson@yahboom:~$ ^[[200~ros2 launch yahboomcar_nav roda_save_map_launch.py map_name:=map~^C  
jetson@yahboom:~$ ros2 launch yahboomcar_nav roda_save_map_launch.py map_name:=map  
[INFO] [launch]: All log files can be found below /home/jetson/.ros/log/2025-12-22-16-35-58-531203-yahboom-995528  
[INFO] [launch]: Default logging verbosity is set to INFO  
[INFO] [map_saver_cli-1]: process started with pid [995561]  
[map_saver_cli-1] [INFO] [1766392558.708373651] [map_saver]:  
[map_saver_cli-1] map_saver lifecycle node launched.  
[map_saver_cli-1] Waiting on external lifecycle transitions to activate  
[map_saver_cli-1] See https://design.ros2.org/articles/node_lifecycle.html for more information.  
[map_saver_cli-1] [INFO] [1766392558.708749021] [map_saver]: Creating  
[map_saver_cli-1] [INFO] [1766392558.709233547] [map_saver]: Configuring  
[map_saver_cli-1] [INFO] [1766392558.721788935] [map_saver]: Saving map from 'map' topic to '/home/jetson/map/map' file  
[map_saver_cli-1] [WARN] [1766392558.739056038] [map_io]: Image format unspecified. Setting it to : pgm  
[map_saver_cli-1] [INFO] [1766392558.739198634] [map_io]: Received a 244 X 117 map @ 0.05 m/pix  
[map_saver_cli-1] [INFO] [1766392558.755578673] [map_io]: Writing map occupancy data to /home/jetson/map/map.pgm  
[map_saver_cli-1] [INFO] [1766392558.756795987] [map_io]: Writing map metadata to /home/jetson/map/map.yaml  
[map_saver_cli-1] [INFO] [1766392558.757045273] [map_io]: Map saved  
[map_saver_cli-1] [INFO] [1766392558.757067226] [map_saver]: Map saved successfully  
[map_saver_cli-1] [INFO] [1766392558.759976907] [map_saver]: Destroying  
[INFO] [map_saver_cli-1]: process has finished cleanly [pid 995561]  
jetson@yahboom:~$
```

- Save the cartographer pose map

#ORIN

```
ros2 service call /write_state cartographer_ros_msgs/srv/WriteState "{filename:  
'/home/jetson/map/map.pbstream'}"
```

#RDK

```
ros2 service call /write_state cartographer_ros_msgs/srv/WriteState "{filename:  
'/home/sunrise/map/map.pbstream'}"
```

```
jetson@yahboom:~$ ros2 service call /write_state cartographer_ros_msgs/srv/WriteState "{filename: '/home/jetson/map/yahboomcar.pbstream'}"  
requester: making request: cartographer_ros_msgs.srv.WriteState_Request(filename='/home/jetson/map/yahboomcar.pbstream', include_unfinished_submaps=False)  
response:  
cartographer_ros_msgs.srv.WriteState_Response(status=cartographer_ros_msgs.msg.StatusResponse(code=0, message="State written to '/home/jetson/map/yahboomcar.pbstream'.") )  
jetson@yahboom:~$
```

- Map files are saved by default in the `~/map` path. You can also specify the map name using the `map_name` parameter when saving the map.

```
jetson@yahboom:~$ ll ~/map/
total 9980
drwxrwxrwx  3 root  root    4096 Dec 23 10:37 ./
drwxr-x--- 51 jetson jetson  4096 Dec 23 10:13 ../
-rw-rw-r--  1 jetson jetson 254907 Dec 22 16:36 map.data
-rwxrwxrwx  1 root  root    10965 Dec 22 17:45 map.geojson*
-rw-rw-r--  1 jetson jetson 2683972 Dec 22 17:31 map.pbstream
-rw-rw-r--  1 jetson jetson  54773 Dec 22 17:25 map.pgm
-rw-rw-r--  1 jetson jetson 7186043 Dec 22 16:36 map.posegraph
-rw-rw-r--  1 jetson jetson    122 Dec 22 17:25 map.yaml
drwxrwxr-x  2 jetson jetson  4096 Dec 16 20:28 text/
```